

Experiment 5: Missionaries and Cannibals Problem

Aim

To solve the Missionaries and Cannibals problem using Python.

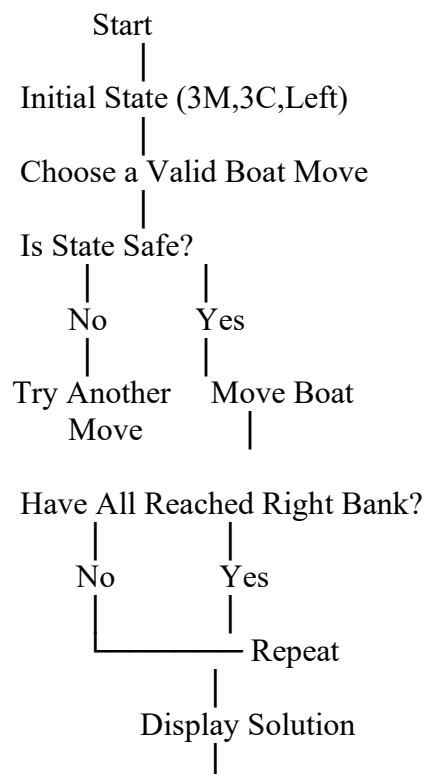
Objective

To safely move all the missionaries and cannibals from the left river bank to the right river bank without leaving the missionaries outnumbered by cannibals on either side.

Algorithm

1. Start with 3 missionaries and 3 cannibals on the left bank.
2. The boat can carry a maximum of 2 people.
3. Move missionaries and/or cannibals across the river.
4. Ensure that on both banks, missionaries are never fewer than cannibals (unless there are no missionaries on that bank).
5. Repeat the valid moves until all six people reach the right bank.
6. Display the sequence of states.
7. Stop.

Flowchart

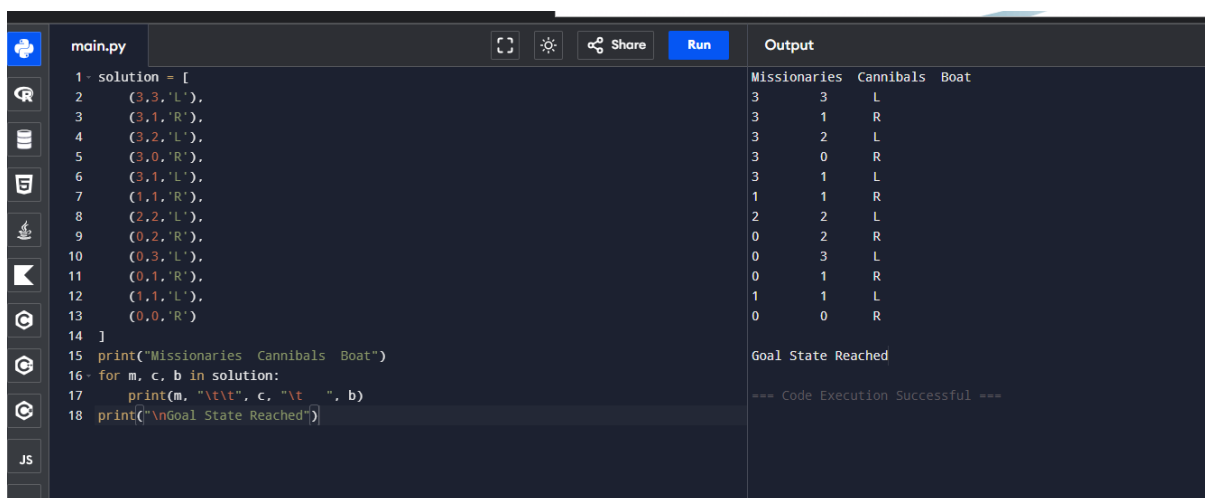


Stop

Python Program

```
solution = [  
    (3,3,'L'),  
    (3,1,'R'),  
    (3,2,'L'),  
    (3,0,'R'),  
    (3,1,'L'),  
    (1,1,'R'),  
    (2,2,'L'),  
    (0,2,'R'),  
    (0,3,'L'),  
    (0,1,'R'),  
    (1,1,'L'),  
    (0,0,'R')  
]  
  
print("Missionaries Cannibals Boat")  
  
for m, c, b in solution:  
    print(m, "\t\t", c, "\t ", b)  
  
print("\nGoal State Reached")
```

Sample Output



The screenshot shows a Python IDE with a dark theme. The left sidebar contains icons for file explorer, search, and other IDE features. The main editor window displays the Python code from the previous block. The right sidebar shows the output of the program, which is a table of states and a confirmation message.

Missionaries	Cannibals	Boat
3	3	L
3	1	R
3	2	L
3	0	R
3	1	L
1	1	R
2	2	L
0	2	R
0	3	L
0	1	R
1	1	L
0	0	R

Goal State Reached

=== Code Execution Successful ===

Result

Thus, the Missionaries and Cannibals problem was solved successfully using Python.

Conclusion

The program successfully transfers all three missionaries and three cannibals to the opposite river bank while satisfying the safety constraint that missionaries are never outnumbered by cannibals on either bank. This problem demonstrates **state-space search** and **constraint satisfaction** concepts used in Artificial Intelligence.